

Spinal cord stimulation exerts analgesia by activating specific receptor subtypes

Analgesia-Response To: Neurotransmitter	Receptor Subtype	P-SCS	B-SCS	HF-SCS	DRG-S
5-HT	5-HT ₁ :	↔ (Song et al., 2011)			
	5-HT ₂ :	↑ (Barchini et al., 2012)			
	5-HT ₃ :	↑ (Song et al., 2011; Barchini et al., 2012)			
	5-HT ₄ :	↑↑ (Song et al., 2011)			
	5-HT ₆ :	↑ (Song et al., 2011)			
	5-HT ₇ :	↔ (Song et al., 2011)			
		↔ (Song et al., 2011)			
GABA	GABA _A :	↑ (Duggan & Foong, 1985; Song et al., 2011; Barchini et al., 2012)			
	GABA _B :	↔ (Cui et al., 1996)			
		↑↑ (Song et al., 2009; Song et al., 2011; Cui et al., 1996; Barchini et al., 2012)			
NE	α1 adrenergic:	↑ (Barchini et al., 2012)			
	α2 adrenergic:	↑↑ (Barchini et al., 2012; Schechtmann et al., 2004)			
	β1 adrenergic:	↑ (Barchini et al., 2012)			
	β2 adrenergic:	↑ (Barchini et al., 2012)			
Dopamine	D2	↑ (Barchini et al., 2012)			
	D3	↑ (Barchini et al., 2012)			
ACh	M1 mAChR:	↑ (Schechtmann et al., 2008)			
	M2 mAChR:	↑ (Schechtmann et al., 2008; Song et al., 2008)			
	M3 mAChR:	↔ (Schechtmann et al., 2008)			
	M4 mAChR:	↑↑ (Schechtmann et al., 2008; Song et al., 2008)			
	nAChR:	↔ (Song et al., 2009)			
		↔ (Schechtmann et al., 2008)			

Activation of specific receptor subtypes contributes to P-SCS mediated analgesia. Experiments used receptor antagonists and agonists to clarify the role of particular receptors. HT3 receptor, GABA_B receptor, α2 adrenergic receptor and M4 mAChR pathways were found to contribute to this analgesia to a greater extent than other receptor subtypes.